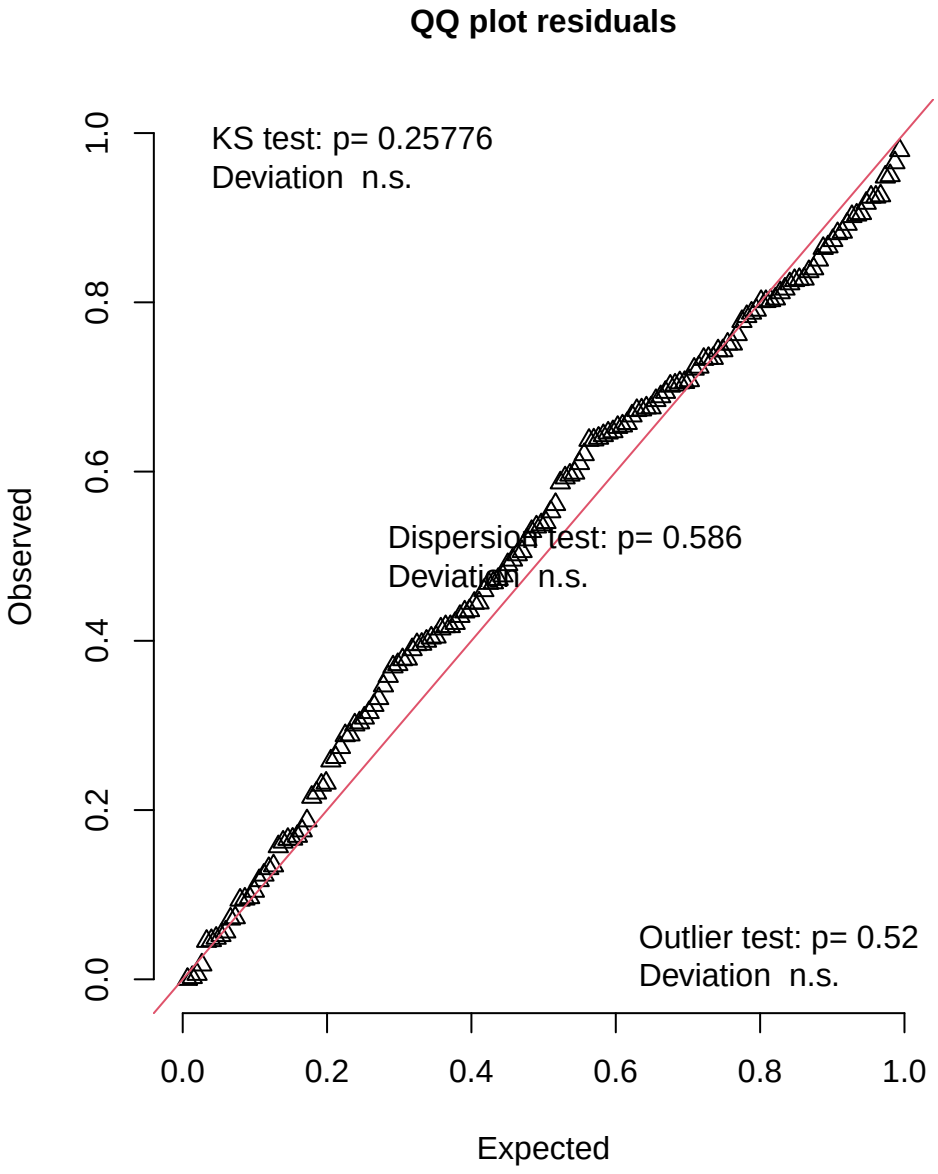
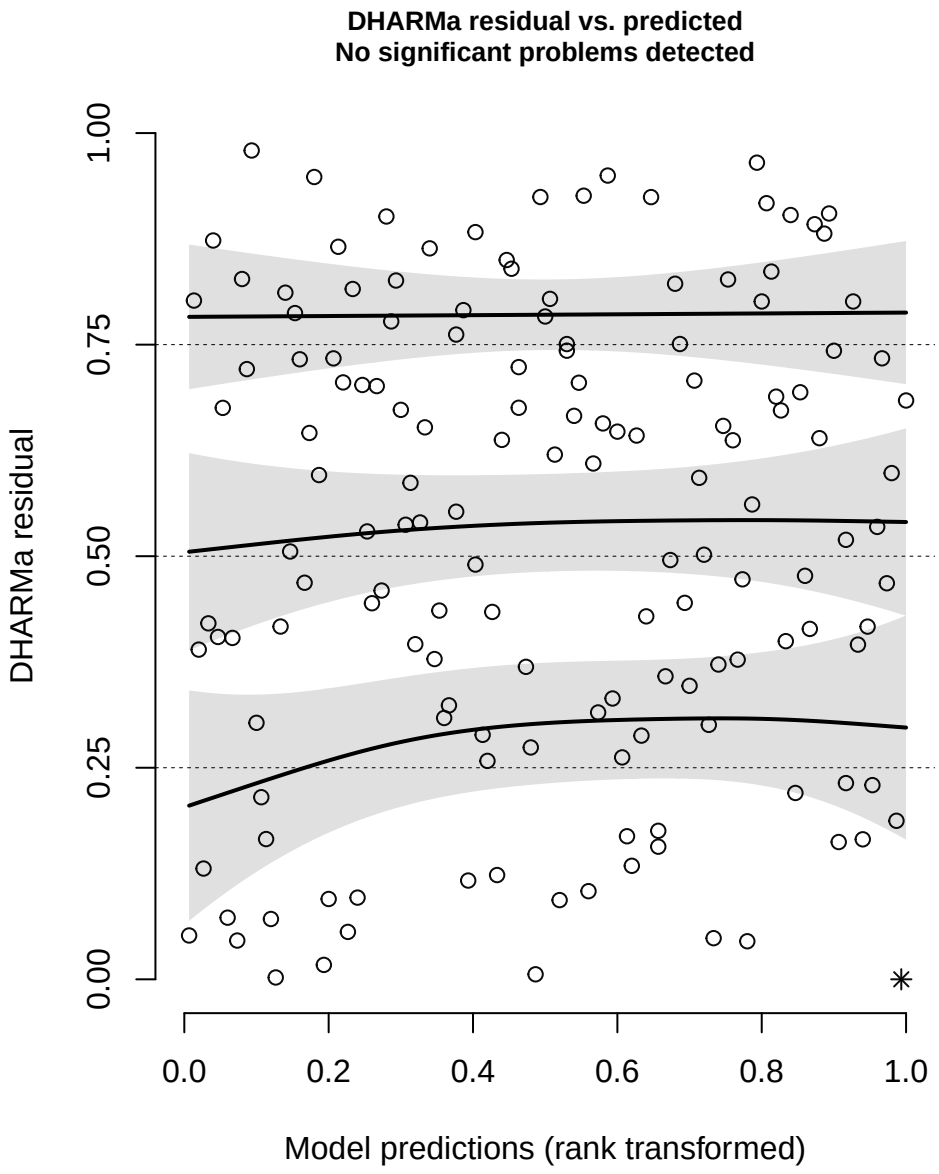
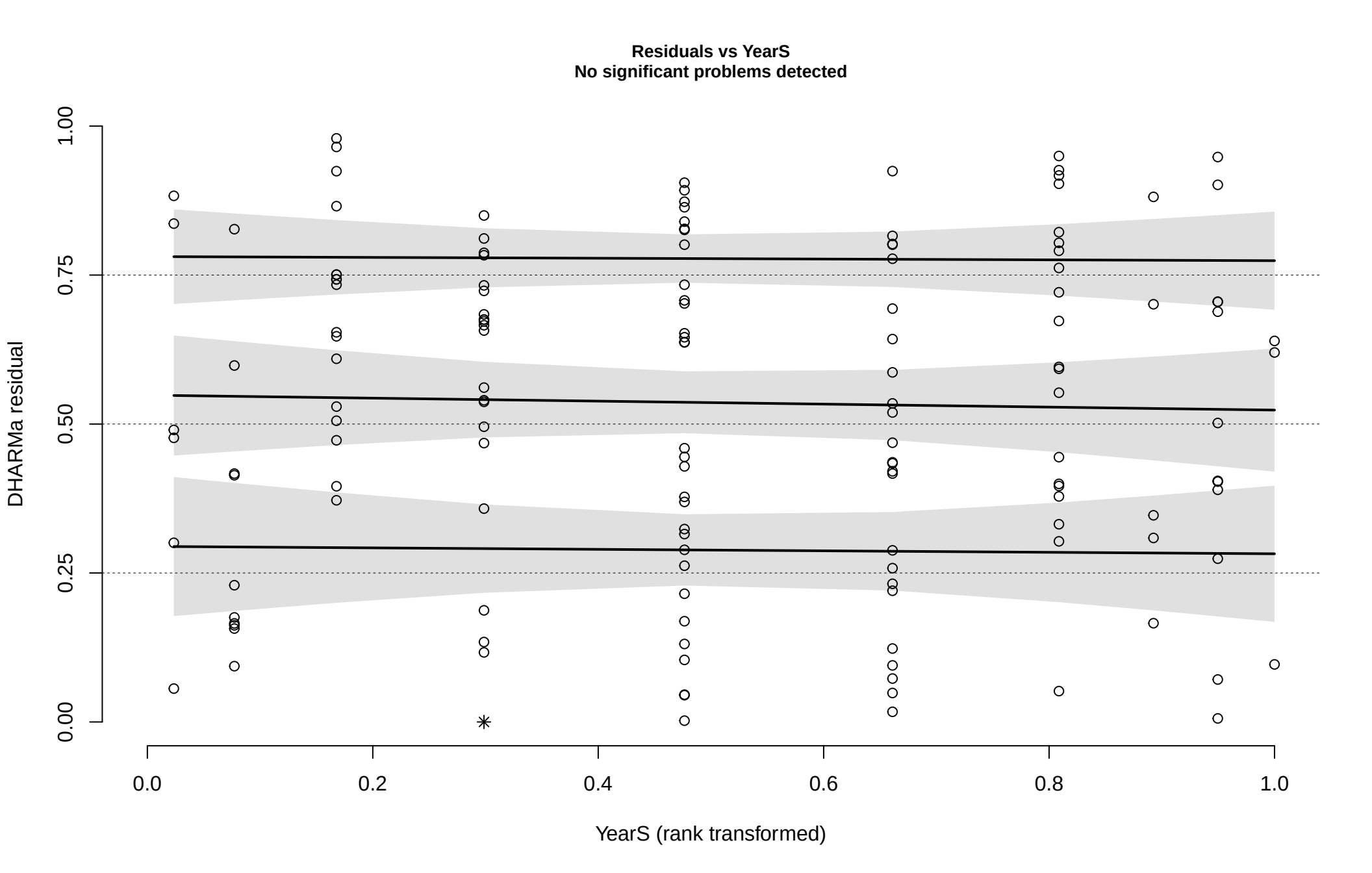
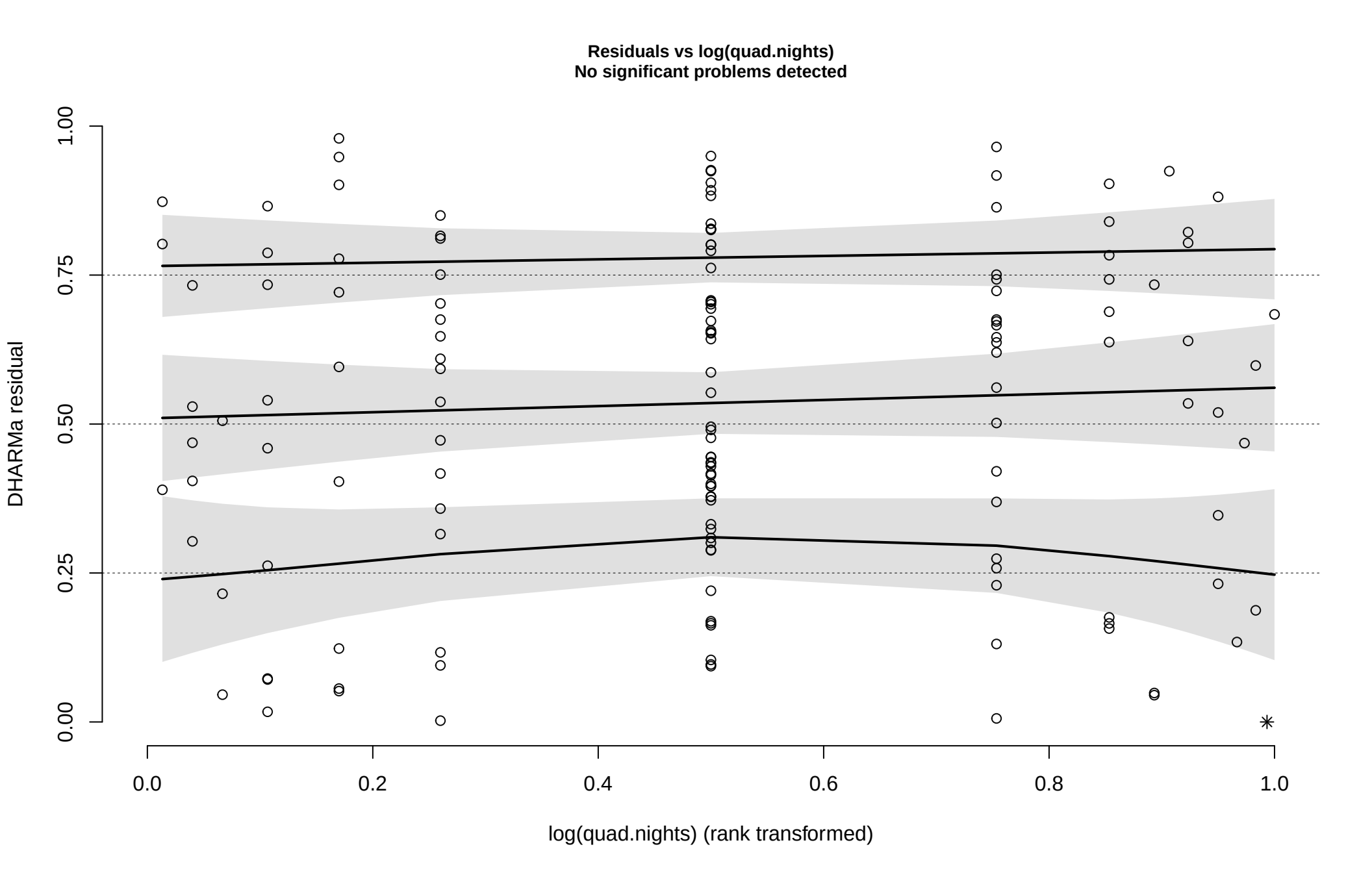
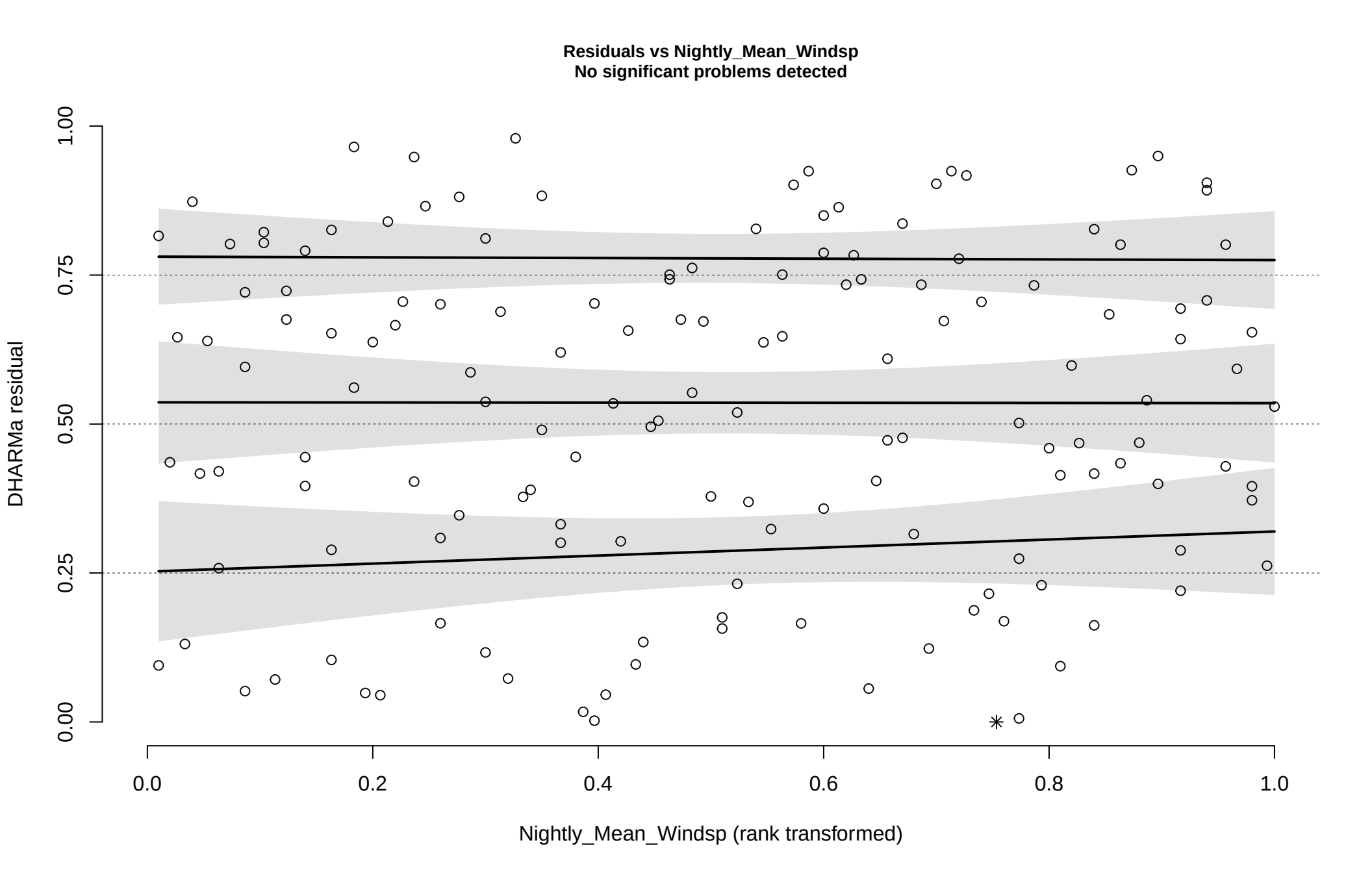
DHARMa residuals: Northern Region_MYVO

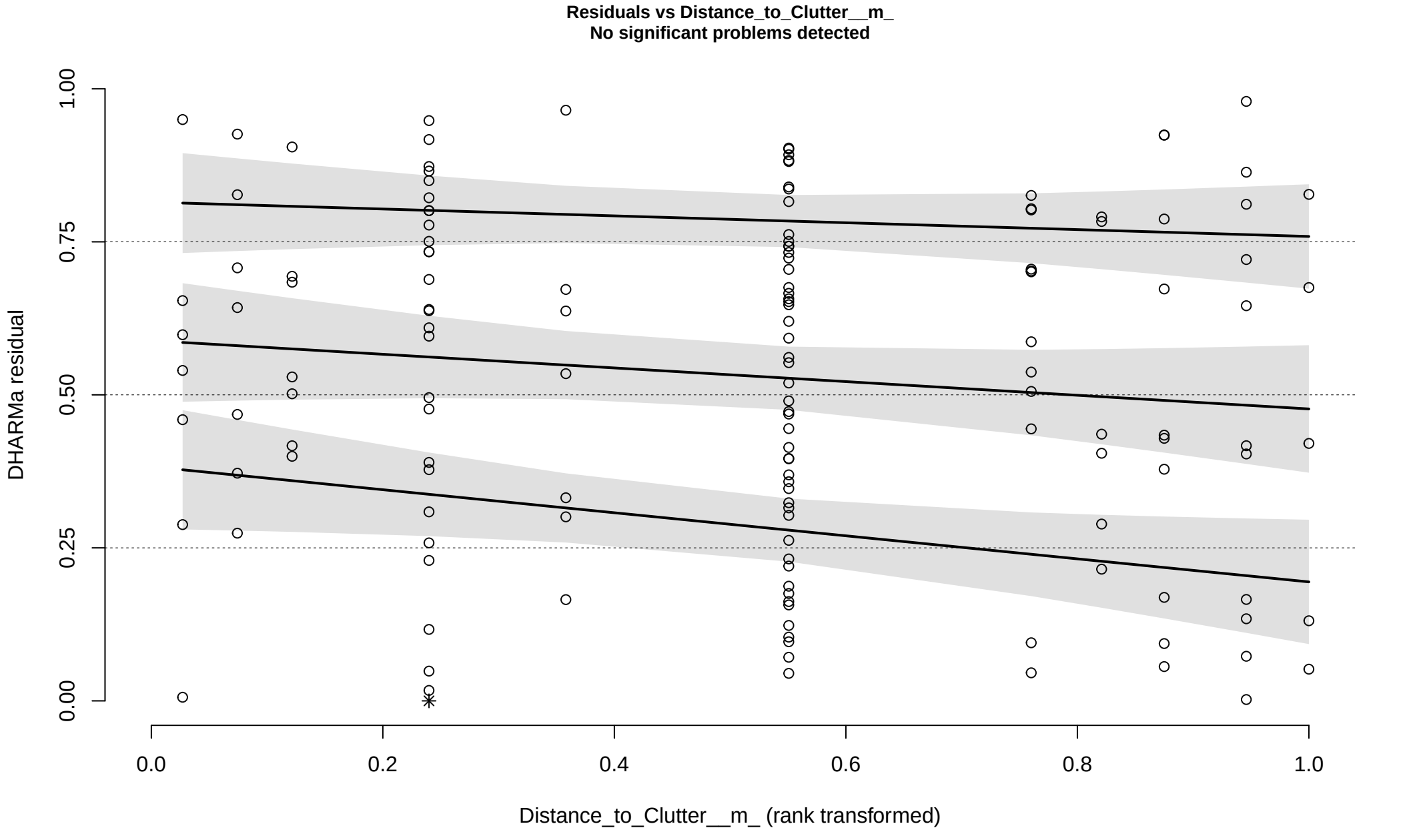


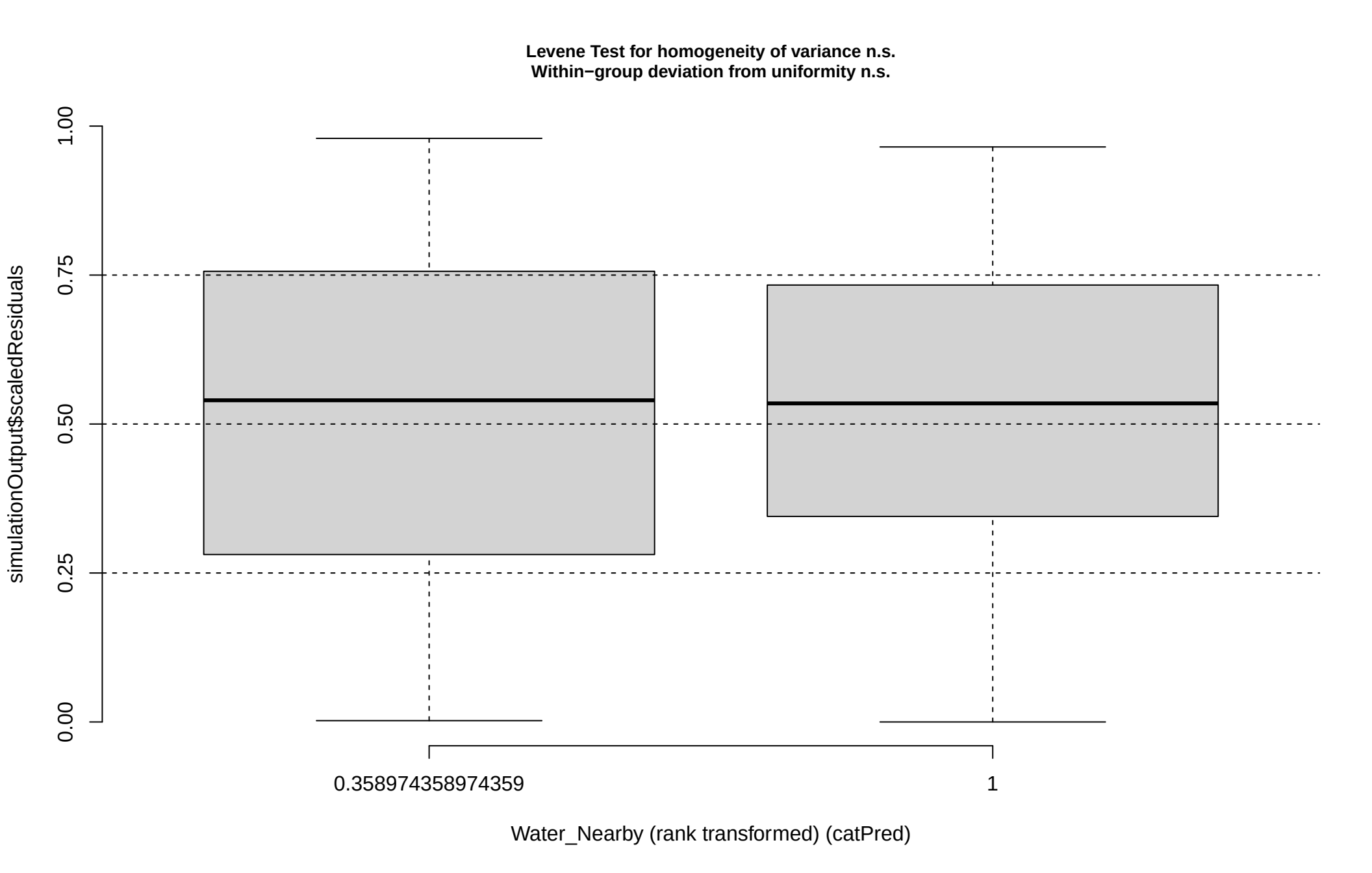


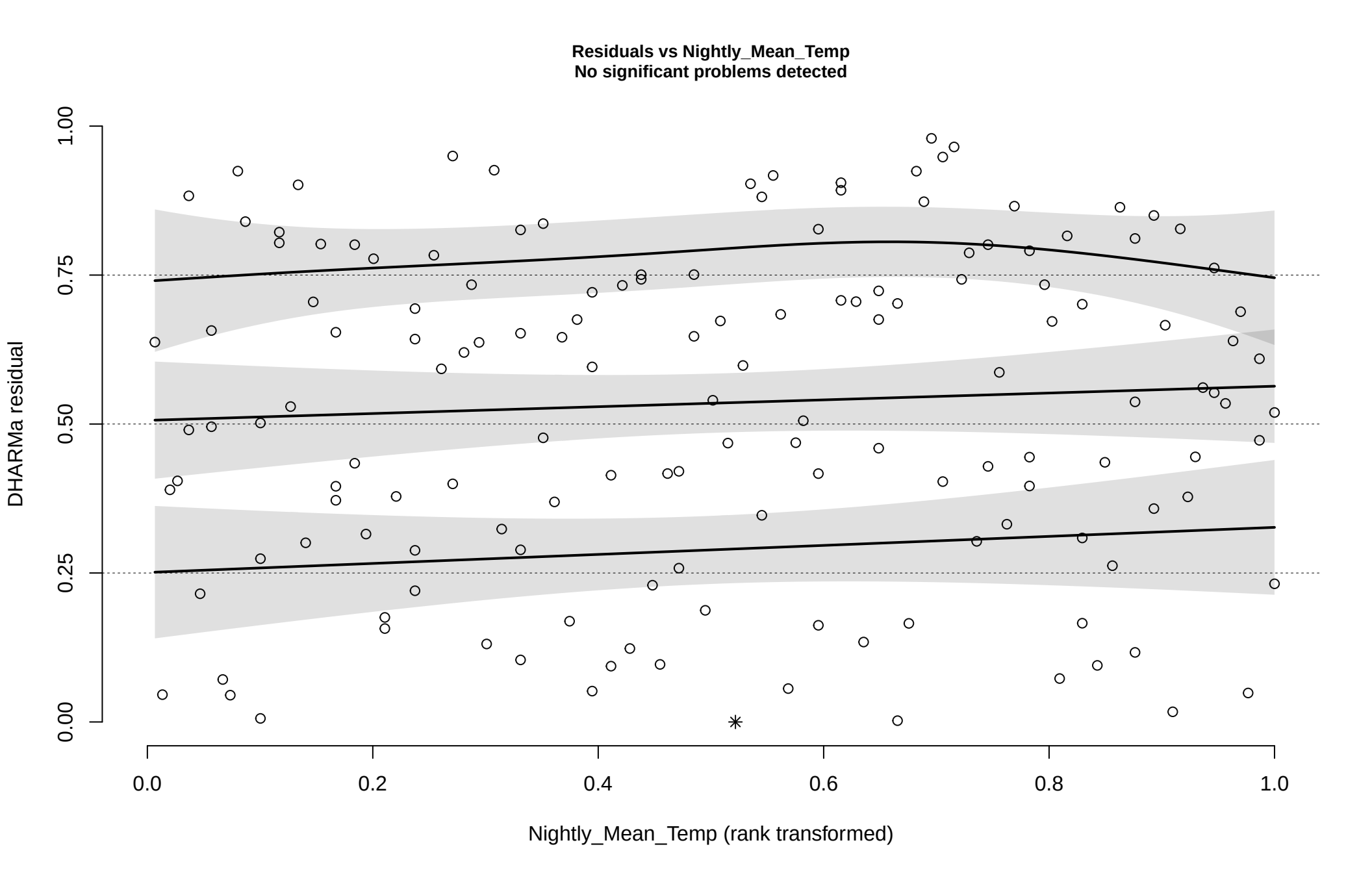

Residuals vs YearS
No significant problems detected
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
YearS (rank transformed)

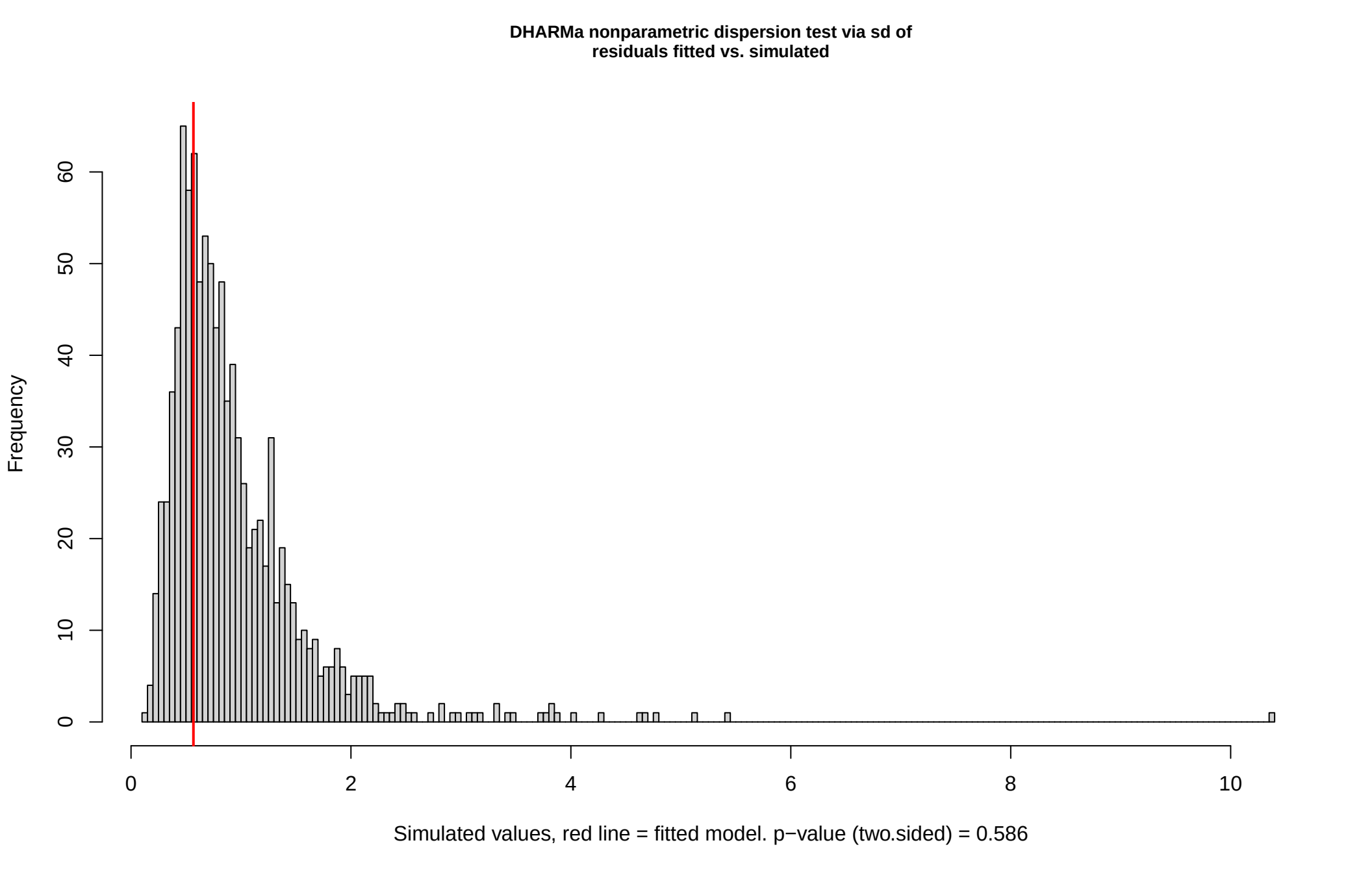

Residuals vs log(quad.nights)
No significant problems detected
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
log(quad.nights) (rank transformed)

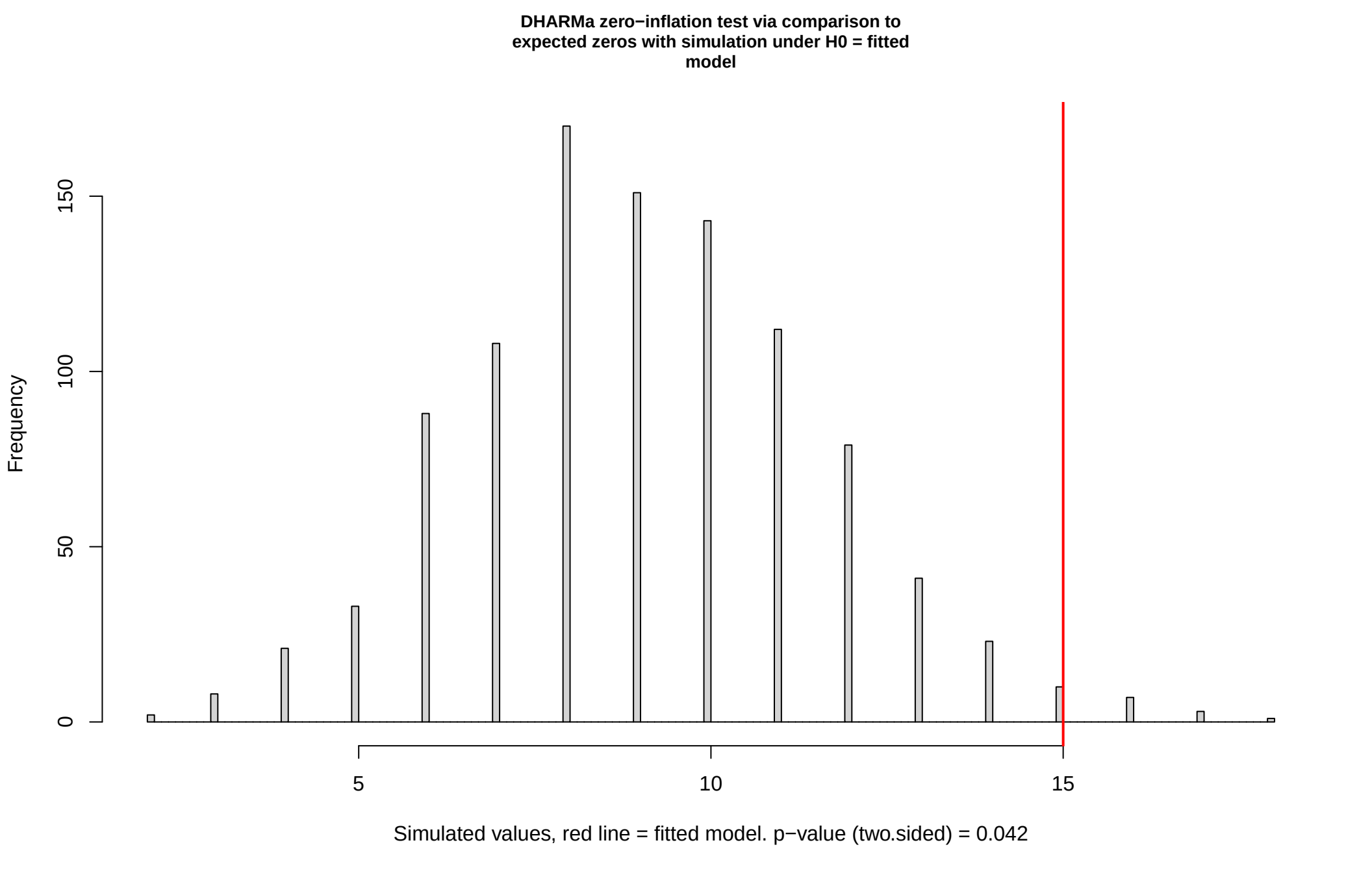

Residuals vs Nightly_Mean_Windsp
No significant problems detected
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_Windsp (rank transformed)

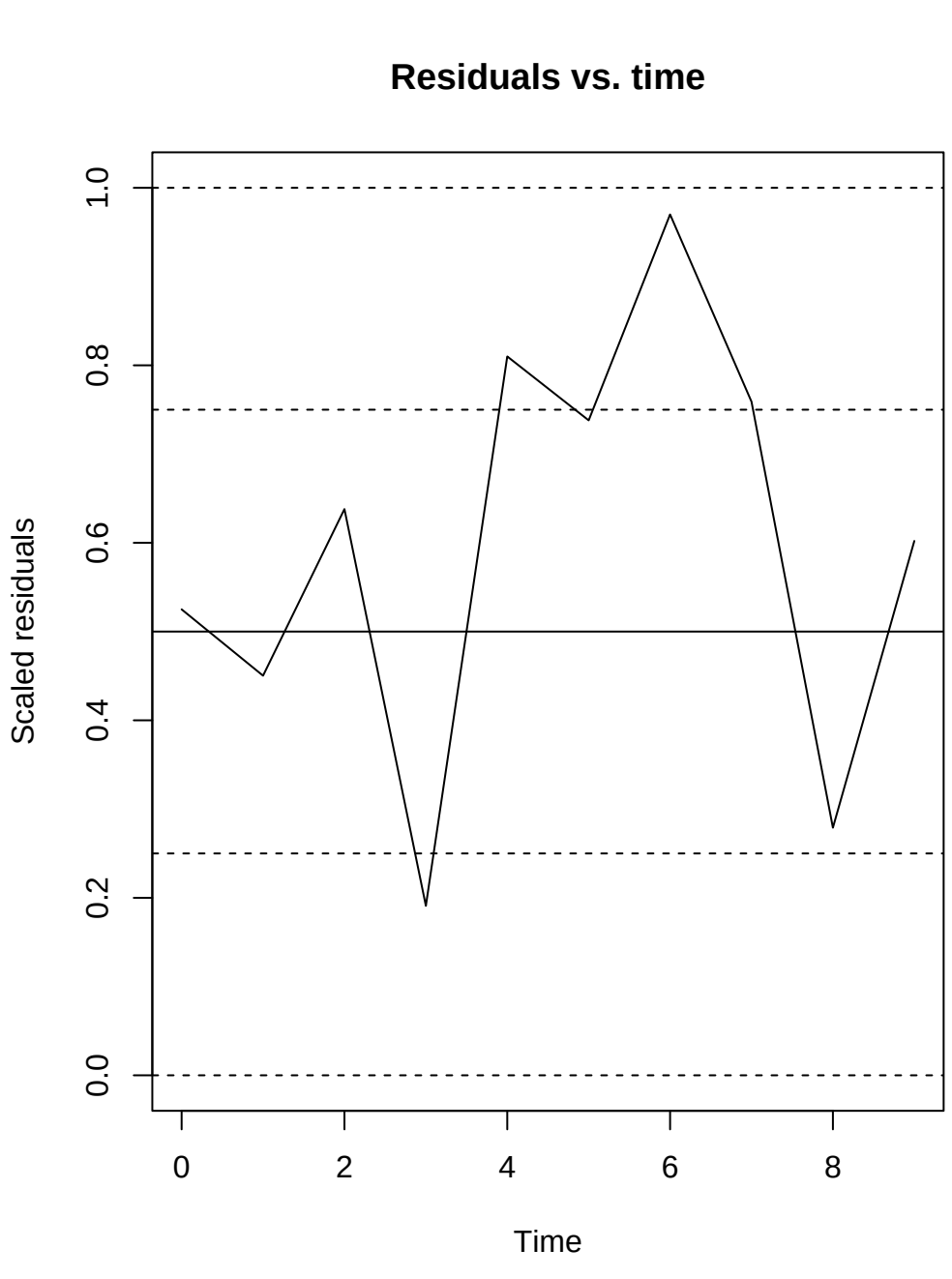

Residuals vs Distance_to_Clutter__m_
No significant problems detected
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Distance_to_Clutter__m_ (rank transformed)

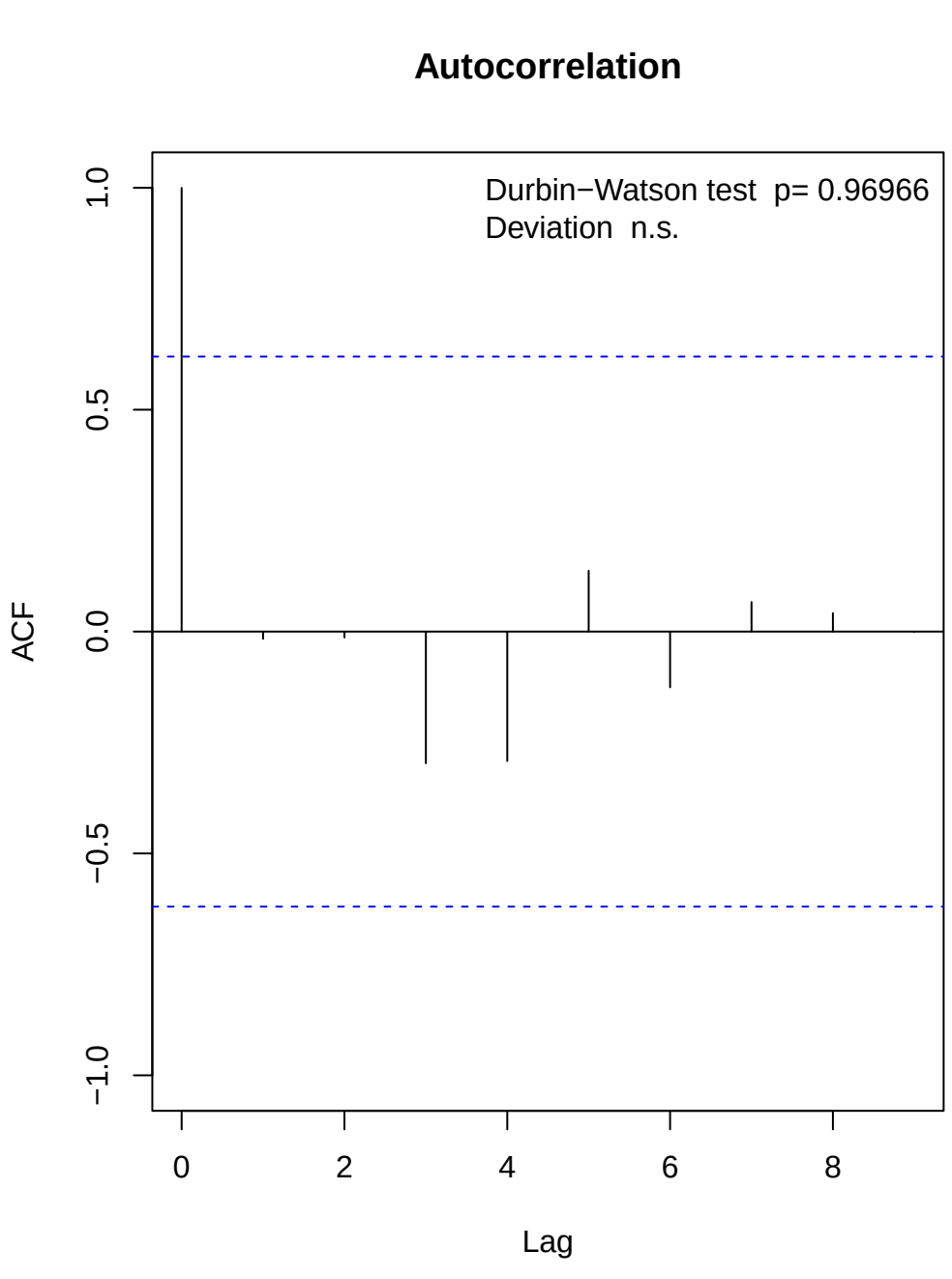

Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
1.00
0.75
0.50
0.25
0.00
0.358974358974359
1
Water_Nearby (rank transformed) (catPred)

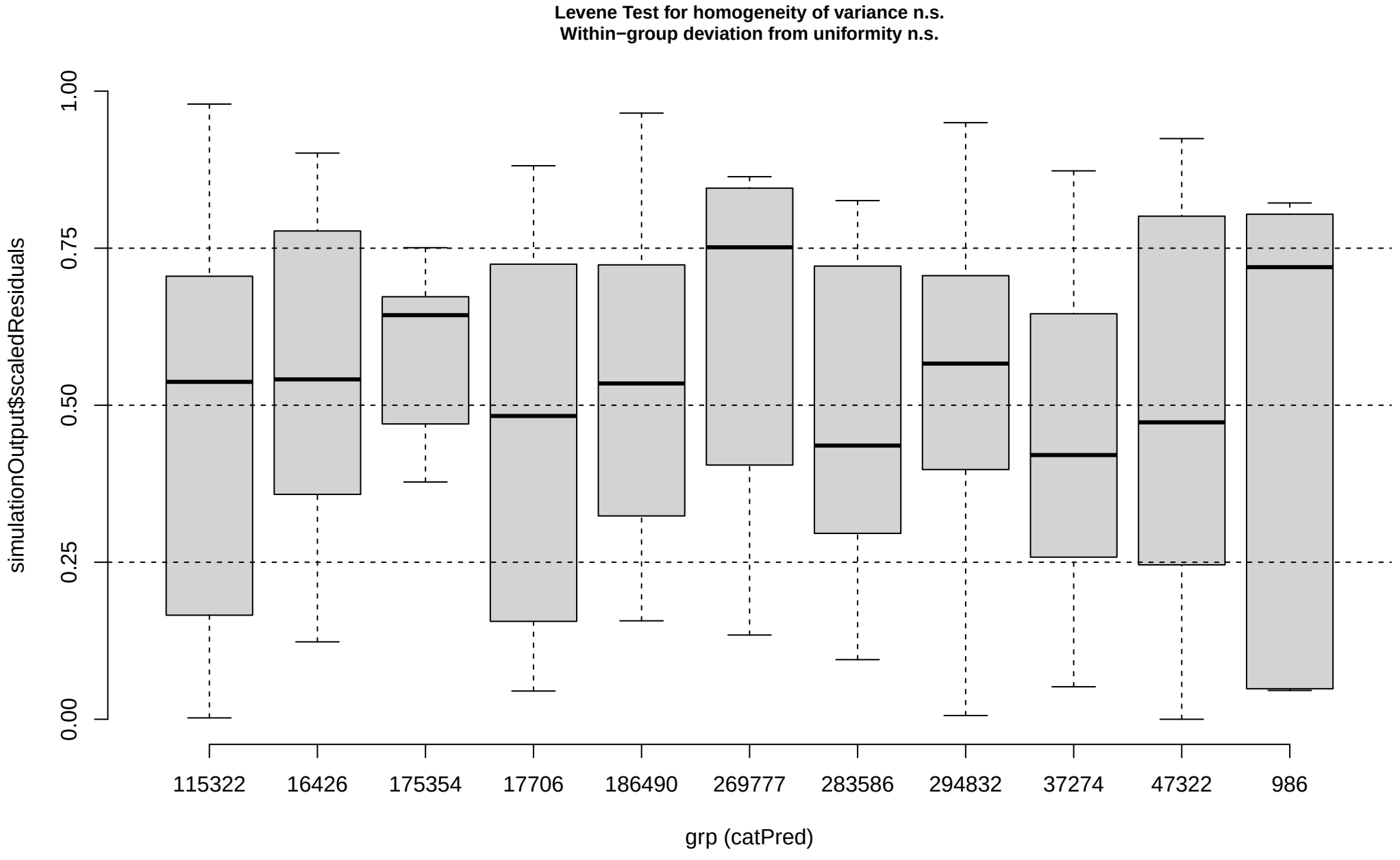

Residuals vs Nightly_Mean_Temp
No significant problems detected
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_Temp (rank transformed)

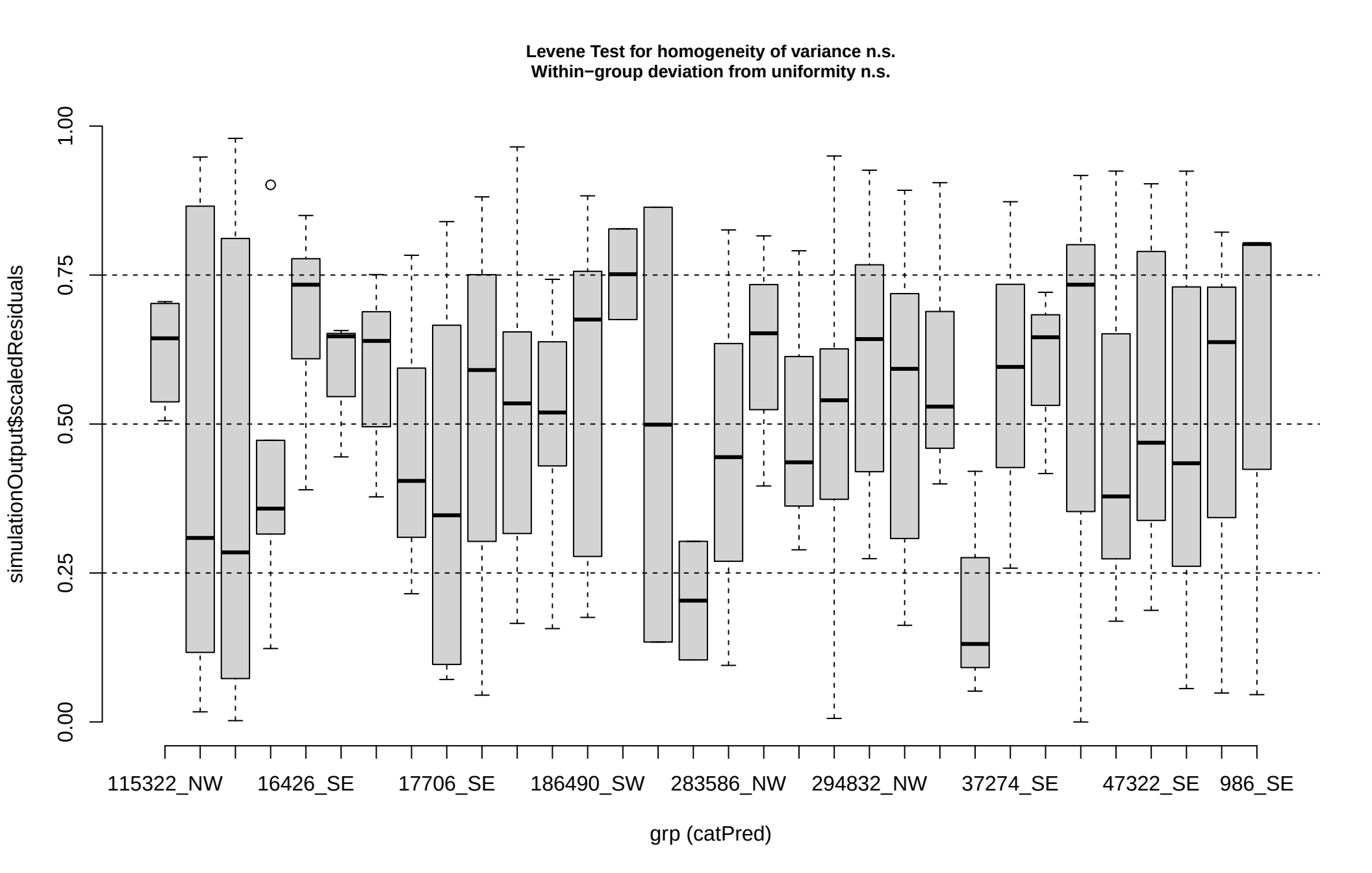

DHARMa nonparametric dispersion test via sd of
residuals fitted vs. simulated
Frequency
0
10
20
30
40
50
60
0
2
4
6
8
10
Simulated values, red line = fitted model. p−value (two.sided) = 0.586

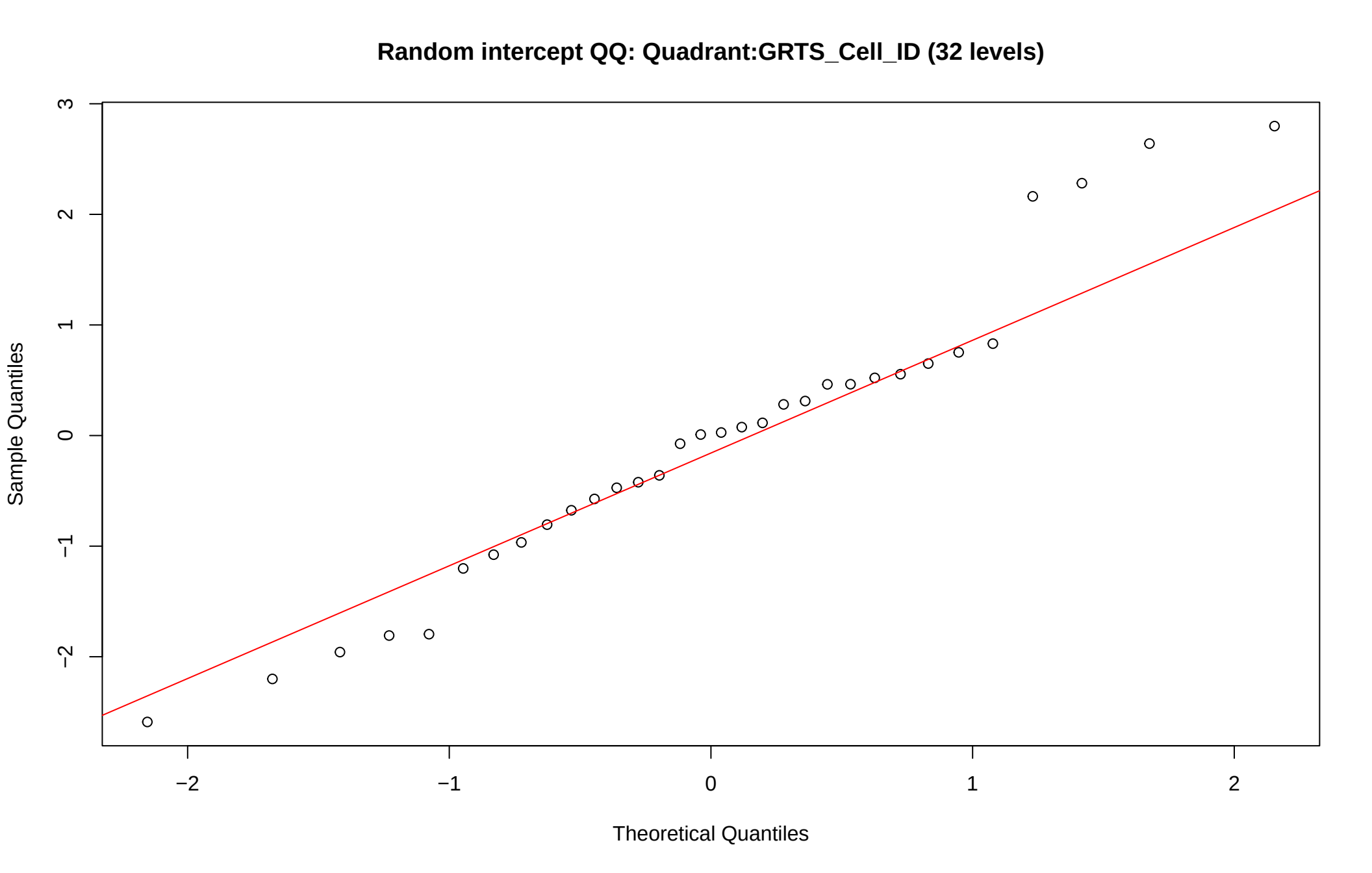

DHARMa zero−inflation test via comparison to expected zeros with simulation under H0 = fitted model
Frequency
0
50
100
150
5
10
15
Simulated values, red line = fitted model. p−value (two.sided) = 0.042

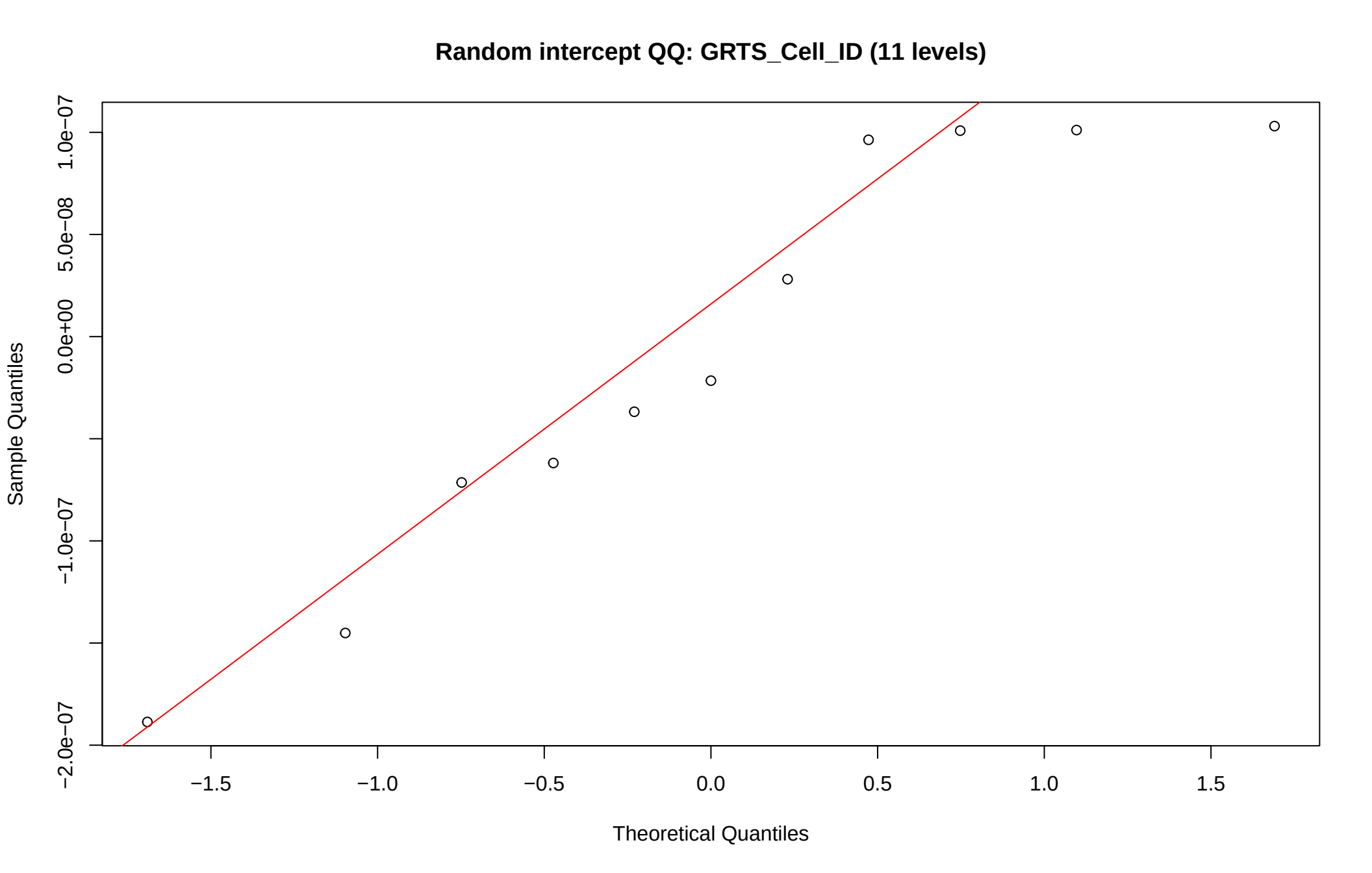

Residuals vs. time
Scaled residuals
0.0
0.2
0.4
0.6
0.8
1.0
0
2
4
6
8
Time


Autocorrelation
Durbin−Watson test p= 0.96966
Deviation n.s.
ACF
−1.0
−0.5
0.0
0.5
1.0
0
2
4
6
8
Lag


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
1.00
0.75
0.50
0.25
0.00
115322
16426
175354
17706
186490
269777
283586
294832
37274
47322
986
grp (catPred)


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
1.00
0.75
0.50
0.25
0.00
115322_NW
16426_SE
17706_SE
186490_SW
283586_NW
294832_NW
37274_SE
47322_SE
986_SE
grp (catPred)


Random intercept QQ: Quadrant:GRTS_Cell_ID (32 levels)
Sample Quantiles
Theoretical Quantiles
3
2
1
0
−1
−2
−2
−1
0
1
2


Random intercept QQ: GRTS_Cell_ID (11 levels)
Sample Quantiles
Theoretical Quantiles
1.0e−07
5.0e−08
0.0e+00
−1.0e−07
−2.0e−07
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5